Mutations of COMP cause two clinically distinct disorders: pseudoachondroplasia and multiple epiphyseal dysplasia. Both are inherited as autosomal dominant disorders, have onset after birth, and are dominated by osteoarthritis of hips and knees. Dwarfism is severe and skeletal deformities are common in pseudoachondroplasia.

Cartilage collagens and COMP are multimeric molecules, that is, they are composed of multiple subunits, three for collagens and five for COMP. Like a square wheel on a car, the products of mutant alleles interfere functionally with the products of normal alleles when they combine during molecular assembly, a so-called dominant negative effect. Most collagen mutations are thought to act through this mechanism to reduce the number of collagen molecules in cartilage matrix, which in turn alters the ability of cartilage to function as a template for bone growth.

Similar types of mutations occur in genes encoding type I collagen, which is the principal matrix protein of bone. These mutations lead to osteogenesis imperfecta (OI), which is a spectrum of disorders of varying severity. The hallmark of OI is bone fractures, although patients often have blue sclerae (the "whites" of the eye), fragile skin, and dental problems that reflect the widespread distribution of type I collagen in many connective tissues.

Excessive Growth

Genetic growth disorders also include conditions with excessive growth. Beckwith-Wiedemann syndrome is characterized by an enlarged tongue, abdominal wall defects (omphalocele), and generalized overgrowth during the fetal and neonatal period. Most of the findings can be attributed to the excess availability of insulin-like growth factor II (IGF2) that results from duplication, loss of heterozygosity, or disturbed imprinting of the IGF2 gene. The syndrome behaves as an autosomal dominant trait in many families. The excessive growth slows with age, but patients are predisposed to childhood tumors, especially **Wilms tumor**.

Wilms tumor a cancerous cell mass of the kidney

Simpson-Golabi-Behmel syndrome is an X-linked overgrowth syndrome with many of the features of Beckwith-Wiedemann syndrome. It results from mutations of glypican 3, which is a cell surface proteoglycan that binds and may sequester growth factors such as IGF2. Glypican 3 mutations appear to enhance IGF2 signaling through its receptor, explaining the clinical similarities between the two syndromes. SEE ALSO Birth Defects; Disease, Genetics of; Genetic Counseling; Hormonal Regulation; Imprinting; Inheritance Patterns; Signal Transduction.

William Horton

Bibliography

Karsenty, G., and E. F. Wagner. "Reaching a Genetic and Molecular Understanding of Skeletal Development." *Developmental Cell* 2, no. 4 (2002): 389–406.

MacGillivray, M. H. "The Basics for the Diagnosis and Management of Short Stature: A Pediatric Endocrinologist's Approach." *Pediatric Annual* 29 (Sept., 2000): 570–575.

Wagner, E. F., and G. Karsenty. "Genetic Control of Skeletal Development." *Current Opinion in Genetic Development* 5 (Oct., 2001): 527–532.

Hardy-Weinberg Equilibrium

H

The Hardy-Weinberg equilibrium is the statement that **allele** frequencies in a population remain constant over time, in the absence of forces to change them. Its name derives from Godfrey Hardy, an English mathematician, and Wilhelm Weinberg, a German physician, who independently formulated it in the early twentieth century. The statement and the set of assumptions and mathematical tools that accompany it are used by population geneticists to analyze the occurrence of, and reasons for, changes in allele frequency. Evolution in a population is often defined as a change in allele frequency over time. The Hardy-Weinberg equilibrium, therefore, can be used to test whether evolution is occurring in populations.

allele a particular form of a gene

Basic Concepts

A population is a set of interbreeding individuals all belonging to the same species. In most sexually reproducing species, including humans, each organism contains two copies of virtually every gene—one inherited from each parent. Any particular gene may occur in slightly different forms, called alleles. An organism with two identical alleles is called homozygous for that gene, and one with two different alleles is called heterozygous. During the formation of **gametes**, the two alleles separate into different gametes. Mating unites egg and sperm, so that the offspring obtains two alleles for each gene.

gametes reproductive cells, such as sperm or eggs

The two alleles for a gene typically have different effects on the phenotype, or characteristics, of the organism. For many genes, one allele will control the phenotype if it is present in either one or two copies; this allele, which is often represented by a single, uppercase letter—*B*, for example—is said to be dominant. The other allele will only exert a visible effect if the dominant allele is not present; it is said to be recessive and is often represented by a lowercase letter—*b*, for example. The genotype of an organism specifies both alleles for a particular gene and is often symbolized by pairs of letters, such as *BB*, *Bb*, or *bb*, with each letter representing an allele.

It is important to understand that "dominant" does not mean an allele is more common in the population—lethal dominant alleles are very rare, for instance. Nor does dominant necessarily mean an allele will spread through the population. Likewise, "recessive" does not necessarily mean an allele will become less common. Indeed, the Hardy-Weinberg equilibrium shows conditions under which allele frequencies remain unaltered over generations.

Assumptions of the Hardy-Weinberg Model

Before examining the mathematical model underlying the Hardy-Weinberg equilibrium, let us look at the assumptions under which it operates:

1. Organisms reproduce sexually.
2. Mating is random.
3. Population size is very large.
4. Migration in or out is negligible.

5. Mutation does not occur.
6. Natural selection does not act on the alleles under consideration.

While the list appears to be so restrictive that no population can meet its requirements, in fact many do, to a very good first approximation. Even more to the point, variation from the Hardy-Weinberg equilibrium tells a population geneticist that one or more of these assumptions is not being met, thereby providing a clue about the forces at work within the population. Perhaps surprisingly, populations need not be very big to meet the conditions above—populations with as few as one thousand to two thousand individuals can do so.

Allele Frequencies Remain the Same Between Generations

Suppose we want to study the allele frequencies of the gene for coloration in a population of moths. The allele for the dark color pattern, *B*, is dominant to the allele for the light color pattern, *b*. In a certain population, the frequency of *B* is found to be 0.9, and that of b is 0.1 (we will see, below, how to determine these frequencies by studying the moths themselves). This means that 90 percent of all the alleles are *B*, and 10 percent are *b*.

The Hardy-Weinberg equilibrium states that, given the above conditions, allele frequencies will not change from one generation to the next. To show this is true, we need some algebra.

Random mating means each allele has an equal chance of being paired with each other allele. During random mating, the likelihood that a *B* allele from a mother will unite with a *B* allele from a father is given by

$$B \times B = 0.9 \times 0.9 = 0.81.$$

The genotype of this offspring will be *BB*.

Similarly, the likelihoods of other combinations:

$$B \times b = 0.9 \times 0.1 = 0.09 \text{ for genotype } Bb;$$

$$b \times B = 0.1 \times 0.9 = 0.09 \text{ for genotype } Bb;$$

and

$$b \times b = 0.1 \times 0.1 = 0.01 \text{ for genotype } bb.$$

Note that the two *Bb* genotypes are the same. Therefore the frequency of *BB* is 0.81, the frequency of *Bb* is 0.18, and the frequency of *bb* is 0.01. These add to 1, just as we would expect, since they represent all the members of the next generation.

Are the allele frequencies still 0.9 and 0.1? For simplicity, imagine we're looking at one hundred individuals, so that eighty-one are *BB*, eighteen are *Bb*, and one is *bb*. Since each individual has two alleles, there are 200 alleles in all.

The number of *B* alleles is given by $(81 \times 2) + (18 \times 1) = 180$.

The number of *b* alleles is given by $(1 \times 2) + (18 \times 1) = 20$.

By comparing 180 to 20, you can see the frequency of *B* is still 0.9 and that of *b* is still 0.1.

Allele Frequencies Can Be Calculated from Phenotypes

This is all very interesting, but it requires knowing the allele frequencies in a population. The power of the Hardy-Weinberg equilibrium formulas is in their ability to allow us to determine these frequencies by simple observation of the population.

When we see an organism with the dominant phenotype, we do not know whether we are looking at a homozygote (*BB*) or a heterozygote (*Bb*). When we see one with the recessive phenotype, though, we know it has the genotype *bb*. If we determine the proportion of individuals in the population with the *bb* genotype, it is a simple step to calculate the frequency of *b*. Let's work backward through our example above. If we determine that 1 percent of the population is homozygous recessive,

$$bb = 0.01.$$

Since the frequency of *bb* organisms is the product $b \times b$, we can take the square root of this number to get the frequency of *b*:

$$b = (bb)^{1/2} = (0.01)^{1/2}$$

and

$$b = 0.1.$$

Since $B + b$ must equal 1 (assuming there are only two alleles), *B* must be 0.9. With these calculations in hand, we can predict what the frequency for each of the other genotypes should be: *BB* is 0.81, and *Bb* is 0.18.

Departures from Equilibrium Indicate Evolutionary Forces at Work

With these simple tools, we can look at populations to see if they conform to these numerical patterns. If they differ, we seek the reasons for the difference in some violation of the Hardy-Weinberg assumptions. Two processes, natural selection and genetic drift, are the most common and important factors at work in most populations that are not at equilibrium.

For example, suppose we find a population in which the recessive allele frequency is declining over time. We might then investigate whether homozygous recessives are dying earlier. (Many genetic diseases, such as cystic fibrosis, are due to recessive alleles.) This could be due to natural selection, in which those that are better adapted to the environment survive longer and reproduce more frequently.

Or suppose we find a population in which there is a smaller-than-expected number of homozygotes of both types, and a larger number of heterozygotes. This could be due to heterozygote superiority—where the heterozygote is more fit than either homozygote. In humans, this is the case for the allele causing sickle cell disease, a type of hemoglobinopathy.

Nonrandom mating is another potential source of departure from the Hardy-Weinberg equilibrium. Imagine that two alleles give rise to two very different appearances. Individuals may choose to mate with those whose appearance is closest to theirs. This may lead to divergence of the two groups over time into separate populations and perhaps ultimately separation into two species.

CALCULATING ALLELE FREQUENCIES AND GENOTYPES FROM THE OBSERVED FREQUENCY OF HOMOZYGOUS RECESSIVES

B: dominant allele frequency.
b: recessive allele frequency.
b = (observed homozygote recessive frequency)$^{1/2}$.
$B = 1 - b$.
$B \times b$ = expected frequency of heterozygotes in the population.
B^2 = expected frequency of homozygous dominants in the population.

In very small populations, allele frequencies may change dramatically from one generation to the next, due to the vagaries of mate choice or other random events. For instance, half a dozen individuals with the dominant allele may, by chance, have fewer offspring than half a dozen with the recessive allele. This would have little effect in a population of one thousand, but it could have a dramatic effect in a population of twenty. Such changes are known as genetic drift. SEE ALSO Gene Flow; Genetic Drift; Inheritance Patterns; Mutation; Population Bottleneck; Population Genetics.

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Bibliography

Hartl, D. L., and A. G. Clark. *Principles of Population Genetics, 3rd ed.* Sunderland, MA: Sinauer, 1997.

Heatshock Protein *See Chaperones*

Hemoglobinopathies

Hemoglobinopathies are diseases caused by the production of abnormal hemoglobin or by a deficiency of hemoglobin synthesis. Hemoglobin is the protein in red blood cells (erythrocytes) that binds to oxygen, to distribute it throughout the body. The major hemoglobinopathies are sickle cell disease and several forms of thalassemia.

Hemoglobin Structure and Function

In the lungs, where oxygen concentration is high, each hemoglobin molecule can bind with one molecule of oxygen. The erythrocyte containing the hemoglobin then travels through the bloodstream to the body's cells, where oxygen concentration is low, and the hemoglobin releases the oxygen for use by local tissue. It also picks up carbon dioxide, and this waste product is transported back to the lungs, where it can be released and exhaled.

Hemoglobin is made up of heme and globin. Heme is an iron-containing pigment that binds to oxygen. Globin, which holds the heme and influences how easily it stores and releases oxygen, is a protein consisting of two pairs of polypeptide chains. Globin can contain several different types of polypeptide chains, termed alpha, beta, and gamma. Each is coded for by a separate gene. The genes are evolutionarily related, and their differences are the result of ancient mutation events in an ancestral form that gave rise to each modern type.

The type of hemoglobin found in healthy adults contains two alpha chains and two beta chains. This form of hemoglobin is called HbA (hemoglobin A). As discussed below, sickle cell disease is due to mutations in the beta chains in HbA. A fetus or newborn baby does not produce HbA. Instead, it produces fetal hemoglobin, or HbF. Like HbA, fetal hemoglobin contains a pair of alpha chains. But in place of the beta chains, it contains a pair of gamma chains. As infants grow older, their bodies produce less and less HbF and more and more HbA.

The Genetics of Hemoglobinopathies

Each person possesses two copies of the beta globin gene, on separate **homologous** chromosomes. In most people, the two copies are identical. A person with two identical gene copies is said to be homozygous.

In some people, the two beta copies are not identical. These people, who have two different **alleles** of the beta globin gene, are said to be heterozygous. The beta globin allele that leads to sickle cell disease is called the hemoglobin S (HbS) allele.

People who have inherited one HbA and one HbS allele are heterozygous for the beta chain gene. They are said to have the sickle cell trait, but not sickle cell disease. As long as they have one HbA allele, these individuals produce sufficient HbA to remain healthy, and they usually do not have any medical problems, or they experience only very mild symptoms. When both alleles must be abnormal to cause a disease, the condition is said to be recessive. Sickle cell disease is a recessive condition.

Sickle cell disease can occur when two individuals who have the sickle cell trait (they are called carriers) have children. Recall that the two beta chain alleles occur on different chromosomes. These homologous chromosomes separate during **gamete** formation, so that each gamete has a fifty-fifty chance of possessing an HbS allele. There is a one-in-four chance that a child conceived by two carriers will inherit a **recessive**, abnormal allele from each parent, and therefore be **homozygous** for the abnormal allele and develop sickle cell disease.

Homozygous forms of hemoglobinopathy can be very serious. Some cause so much damage that the fetus dies before birth, while others require lifelong treatment.

Hbs (sickling hemoglobin) causes red blood cells to take on a sickle shape under conditions of low oxygen.

homologous similar in structure

alleles particular forms of genes

gamete reproductive cell, such as sperm or egg

recessive requiring the presence of two alleles to control the phenotype

homozygous containing two identical copies of a particular gene

Sickle Cell Disease

Sickle cell disease is the most prevalent genetically based disease in the United States. Approximately 1 in 12 Americans of African descent are carriers, having one allele coding for HbS and one gene for HbA. About 1 in 375 Americans of African descent are homozygous for HbS and have the active disease. High occurrence of the HbS allele also occurs in people who live, or whose ancestors lived, in certain parts of Asia, the Mediterranean, and the Middle East.

The alpha chain gene is found on chromosome 11. Each gene is made up of a very long strand of nucleotides. In sickle cell disease, there is a change in only one **nucleotide** in the sequence that codes for the beta chain: A thymine is substituted for an adenine.

nucleotide the building block of RNA or DNA

Genes code for proteins. Because of that change in one nucleotide, a slightly different protein is produced. HbS differs from HbA by only one amino acid: Glutamic acid in HbA is replaced by valine in the sixth position on the beta chain. The substitution does not affect the hemoglobin molecule's ability to bind with oxygen. HbS can carry oxygen just as effectively as HbA. However, glutamic acid is a hydrophilic ("water-loving") amino acid, whereas valine is hydrophobic ("water-hating"). The valine occurs on the outside of the beta chain. The hydrophobic portions of HbS molecules are attracted to each other. When the concentration of oxygen is low, as it is deep in the body's tissues, HbS molecules will attach to each other. Since a single red blood cell contains about 250 million hemoglobin molecules, this can result in very long chains, or **polymers**.

polymers molecules composed of many similar parts

polymerization linking together of similar parts to form a polymer

The **polymerization** that occurs distorts the red blood cell into a curved, sickle shape. Whereas normal erythrocytes travel smoothly through the blood vessels, these unusually elongated and pointed erythrocytes move much more slowly and can block smaller blood vessels. Both the slow movement and the blockages further reduce the amount of oxygen in the blood, promoting even more polymerization and sickling.

The decreased amount of oxygen in the blood also damages local tissues and will cause permanent damage if it lasts long enough. The lack of oxygen is very painful. This progressive cycle of worsening symptoms, called a vaso-occlusive crisis, can last for more than a week.

People with sickle cell disease often develop other health problems. For example, the crescent shaped erythrocytes have shorter life spans than normally shaped cells do. A healthy red blood cell lives about 120 days, while a sickle cell lives only for 10 to 30 days. The body is unable to replace the red blood cells quickly enough, resulting in anemia.

Situations that cause the body to use up oxygen, such as exercise, can precipitate a vaso-occlusive crisis. Also, because dehydration causes the hemoglobin molecules to be packed more tightly together within the erythrocyte, insufficient fluid intake can also cause red blood cells to sickle.

Treatment Options and Continuing Research

Therapy for sickle cell disease used to focus on easing symptoms and treating infections, which are the most common cause of death in children who have this disease. Newer therapies actually treat the disease.

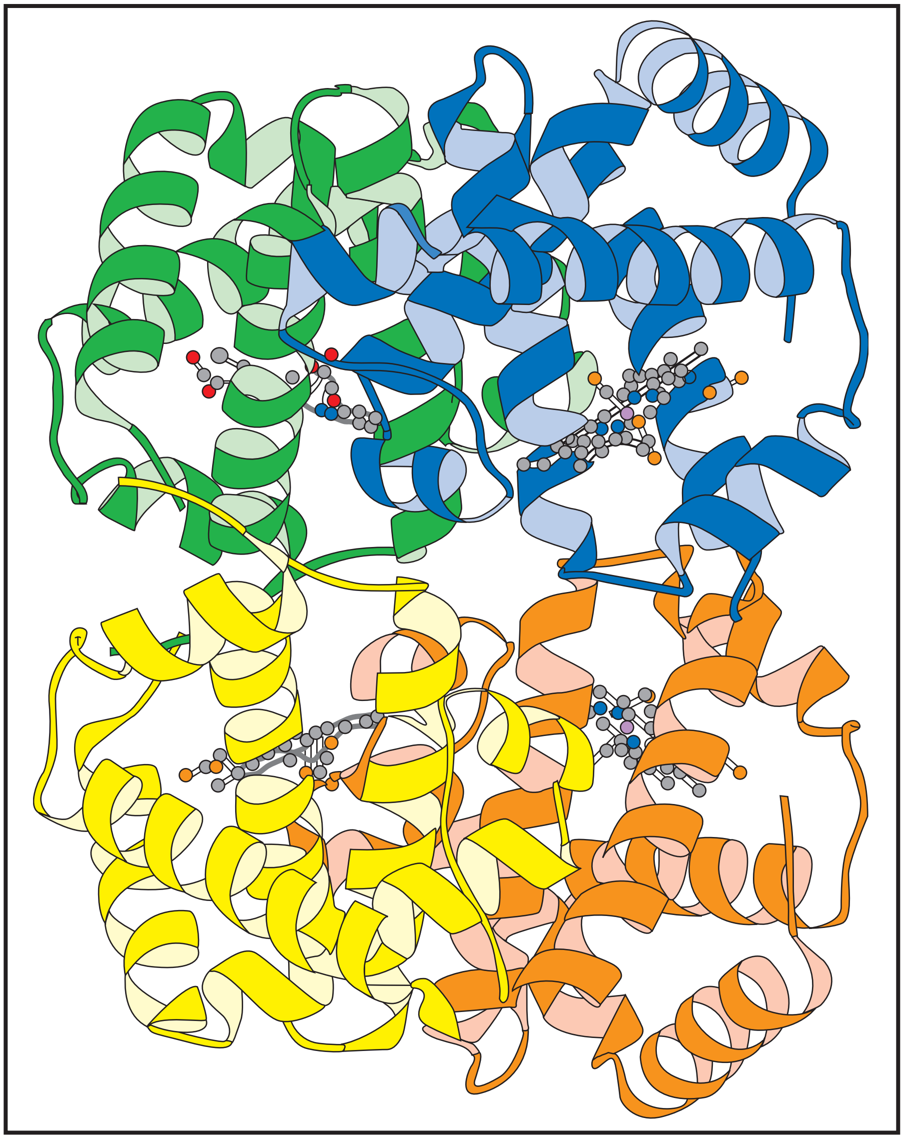
Hydroxyurea and erythropoietin, for example, are two medications that stimulate the bone marrow to produce more fetal hemoglobin, HbF. Production of both red blood cells and hemoglobin occurs in this spongy tissue, which is located in certain bones.

Fetal hemoglobin can transport oxygen but does not polymerize, so the red blood cells cannot sickle. Thus these drugs can prevent vaso-occlusive crises. However, they do have side effects that can limit their usefulness. Hydroxyurea, for example, can suppress bone marrow function.

Normally, the production of HbF is turned off shortly after birth. Scientists are trying to determine how to reactivate the gene for HbF so that the bone marrow of people with sickle cell disease can continually produce fetal hemoglobin without the use of medications. Other research focuses on learning how to insert normal beta chains and regulatory genes into **stem cells**, which are cells that develop into erythrocytes.

stem cells cells capable of differentiating into multiple other cell types

Bone marrow transplants are a new treatment and have largely been conducted in Europe. The donor bone marrow will produce normal hemoglobin and normal red blood cells. However, the tissue must come from an immunologically compatible donor. Also, a bone marrow transplant is a complicated process, and some people have died during the procedure.



A ribbon diagram of hemoglobin. The four ball-and-stick structures are the heme groups. The helices and loops are the two alpha and two beta globin chains. Adapted from <http://www.ee.seikei.ac.jp/user/seiichi/lecture/Biomedical/02/graphics/hemoglobin.gif>.

Hemoglobin C Disease

Hemoglobin C (Hbc), which is also found in people of African or Mediterranean descent, is very similar in structure to HbS. Both are caused by a change in the sixth residue of the beta chain. While valine replaces glutamic acid to form HbS, the amino acid lysine is found in this position, in HbC.

The substitution of lysine does not cause **pathological** changes in the hemoglobin molecule. People who are homozygous for HbC typically have red blood cells that appear unusual, but they do not sickle. These individuals have no symptoms, and they do not require treatment.

pathological disease-causing

Some people have one gene for HbC and another for HbS. They have hemoglobin SC disease, which usually is much less severe than sickle cell disease.

The Thalassemias

The thalassemias are a group of hemoglobinopathies that, like sickle cell disease, are caused by a genetic change. Unlike sickle cell disease, however, the genetic change does not result in the production of an abnormal form of the globin molecule. Instead, the bone marrow synthesizes insufficient amounts of a hemoglobin chain. This, in turn, reduces the production of red blood cells and causes anemia.

Either the alpha or beta chain may be affected, but beta thalassemias are more common. Individuals who are heterozygous for this disorder have one allele for this disease and one normal allele and are said to have thalassemia minor. They usually produce sufficient beta globin so that they have only mild anemia. They may not have any symptoms at all. Thalassemia minor is sometimes misdiagnosed as iron deficiency anemia.

If two individuals with thalassemia minor have children, there is a one-in-four chance that each child will inherit an abnormal gene from both parents and will be homozygous for the disorder.

Individuals who are homozygous for this condition may develop either thalassemia intermedia or thalassemia major. Newborn babies are healthy because their bodies are still producing HbF, which does not have beta chains. During the first few months of life, the bone marrow switches to producing HbA, and symptoms start to appear.

In thalassemia major, also called Cooley's anemia, the bone marrow does not synthesize beta globin at all. Children affected by thalassemia major become very anemic and require frequent blood transfusions. They are so ill that they often die by early adulthood.

In thalassemia intermedia, the production of beta globin is decreased, but not completely. People with this disease have anemia, but they do not require chronic blood transfusions to stay alive.

Alpha thalassemia is more complicated, because an individual inherits two alpha globin genes from each parent for a total of four alpha globin genes. Thus a person can inherit anywhere from zero to four normal genes.

The more abnormal alpha genes that are inherited, the greater the symptoms. If an individual does not have any functional alpha genes, the body cannot produce any alpha globin. Since HbF requires alpha chains, the developing fetus does not produce healthy hemoglobin and shows severe symptoms even before birth. This condition is almost always fatal, with affected infants dying either before or shortly after delivery.

The loss of three functional alpha genes produces severe anemia, the loss of two functional genes typically causes mild anemia, and the loss of only one gene usually does not produce any symptoms. The thalassemias most commonly occur in people from Italy, Greece, the Middle East, Africa, and Southeast Asia; and in their descendants.

Treatment Options and Continuing Research

Blood transfusions have been a common therapy for severe thalassemia, but transfusions do not cure the disease, and frequent transfusions can cause iron overload, an illness caused by excessively high levels of iron. A drug, called an iron chelator, may be given to bind with the excess iron. Iron chelators can produce additional side effects, such as hearing loss and reduced growth.

As with sickle cell disease, gene therapy and bone marrow transplants are very promising therapies for severe thalassemias. While transplants are risky procedures and can cause death, they are more likely to be successful when performed on young and relatively healthy children.

The Heterozygous Advantage

Being homozygous for either sickle cell disease or thalassemia can result in serious illness, but being heterozygous for either condition may actually be beneficial under certain circumstances. Both diseases occur primarily in people who live, or whose ancestors lived, in parts of the world where malaria occurs.

Malaria is spread by a mosquito, but it is caused by plasmodia, single-celled organisms that, during an infection, reproduce inside red blood cells. Before the development of modern sanitation and medicine, malaria was a common cause of death. But people who had either the sickle cell trait or thalassemia minor—people who were heterozygous for either condition—were much more likely to survive an infection than were people homozygous for HbA.

This "heterozygote advantage" meant that these individuals tended to live longer, have children and pass their genes on to the next generation. While some of their children died from thalassemia or sickle cell disease, about half of them were heterozygous and benefited from the heterozygote advantage. This survival advantage explains the high prevalence of these alleles in these populations.

Today, especially in developed countries, there are effective methods for preventing and treating malaria. Nevertheless, the genes for sickle cell disease and thalassemia still exist and are passed down to children who will never be exposed to malaria. It is likely that these genes will very slowly be lost from the gene pool. SEE ALSO GENOTYPE AND PHENOTYPE; HETEROZYGOTE ADVANTAGE; INHERITANCE PATTERNS; MUTATION; POPULATION SCREENING; PROTEINS; TRANSPLANTATION.

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Bibliography

Weatherall, D. J. "ABC of Clinical Haematology: The Hereditary Anaemias." *British Medical Journal* 314 (1997): 492–496.

Internet Resources

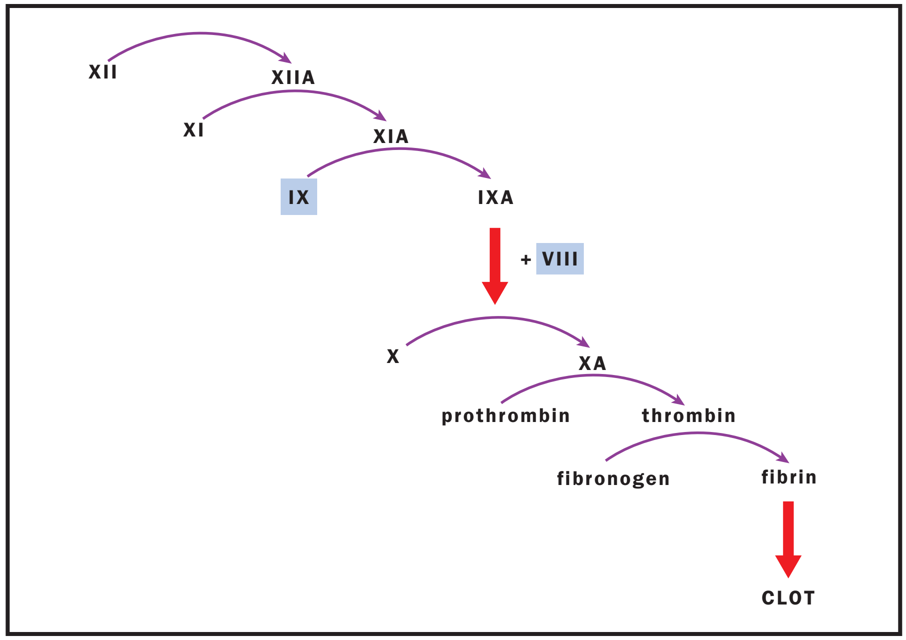
"Bioelectronics Laboratory Index." Seikei University. <http://www.ee.seikei.ac.jp/user/seiichi/lecture/Biomedical/02/graphics/hemoglobin.gif>.

Facts about Sickle Cell Disease. National Heart, Lung, and Blood Institute. <http://www.nhlbi.nih.gov/health/public/blood/sickle/sca_fact.txt>.

Joint Center for Sickle Cell and Thalassemic Disorders. <http://sickle.bwh.harvard.edu>.

Hemophilia

Hemophilia A and hemophilia B are genetic disorders in the blood-clotting system, characterized by bleeding into joints and soft tissues, and by excessive bleeding into any site experiencing trauma or undergoing surgery. Hemophilia A and B are clinically indistinguishable. Both have the same type of bleeding manifestations, and both affect males almost exclusively.



The coagulation cascade involves multistep conversion of several factors into their active (A) forms. In hemophilia, factors VIII or IX are missing or present in insufficient amounts.

The two conditions can be distinguished by detecting the responsible defective proteins.

Contrary to popular belief, individuals with hemophilia rarely bleed excessively from minor cuts or scratches. Hemophilia A and B are both worldwide in distribution, affecting all racial and ethnic groups. The prevalence of hemophilia A is approximately 1 in 10,000 males, and that of hemophilia B is 1 in 30,000 males.

The disorder was recognized (although not named) in Babylonian times. A second-century Jewish rabbi gave permission for a woman not to have her third son circumcised after her first two sons died from the procedure. No doubt the most famous carrier of hemophilia was the nineteenth-century Queen Victoria, whose son, Prince Leopold, had the disorder, and whose two daughters inherited and passed on the gene. Ultimately several of the European royal families were affected by Queen Victoria's gene.

There are a number of clotting factors that interact to form a stable blood clot following injury or surgery. Each factor has a name and is produced in certain cells (usually in the liver), encoded by a certain gene. Hemophilia is caused by a defect in one of these genes. In the case of hemophilia A, factor VIII (FVIII) is deficient or absent. In hemophilia B, FIX is deficient in amount or absent, or it does not function properly.

The genes encoding FVIII and FIX are on the long arm of the X chromosome. Because males have only one X chromosome, hemophilia A and hemophilia B affect males almost exclusively. If a boy's X chromosome has the defective gene that causes hemophilia A or B, the boy will have hemophilia.

Since boys with hemophilia have only an X chromosome carrying the defective gene that causes hemophilia A or B, and no gene for the production of normal FVIII (or FIX), and since fathers pass an X chromosome on to their daughters, all of their daughters will be "obligate carriers" for hemophilia. Obligate carriers—individuals who are definitely carriers—include daughters of men with hemophilia and women who have a maternal family history of hemophilia and one or more affected sons or grandsons.

In general, such carrier females do not bleed excessively, as their other X chromosome, with a gene for normal FVIII (or FIX) production, results in intermediate levels of FVIII (or FIX). However, carrier females have a fifty-fifty chance of passing their X chromosome that bears the hemophilia gene to each child they might have. A son would have an equal chance of being normal or having hemophilia. A girl would have an equal chance of being normal or being a carrier, like her mother.

Gene Defects Causing Hemophilia

Hundreds of defects in the FVIII gene have been shown to cause hemophilia A. These include deletions of varying sizes in the gene, **stop codons**, frameshift mutations, and point mutations. Inversion of the gene is the most common mutation. The same types of defects are found in the FIX gene. This makes population screening for hemophilia impractical: There are too many possible mutations to screen for. However, affected members of a given family will all have the same defect.

stop codons RNA triplet that halts protein synthesis

It is useful to determine (by gene analysis) which defect is present in the FVIII or FIX gene of a particular family with hemophilia, so that one can look for this defect in possible carrier females. Identification of carrier females permits genetic counseling and decision making, on the part of parents, regarding childbearing.

One form of hemophilia is due to the insertion of a transposable genetic element. (DNA sequence that can be copied and moved in the genome).

Detection of FVIII or FIX Gene Defect in Family: Carrier Detection

A number of different techniques are available for carrier detection. Linkage analysis using DNA **polymorphisms** to track defective FVIII or FIX genes is possible in large families. Use of intragenic polymorphisms in both the FVIII and FIX genes allows precise detection of carrier females in most families studied. In persons with hemophilia A, the FVIII gene inversion can be tested for by **Southern blotting**.

polymorphisms occurring in several forms

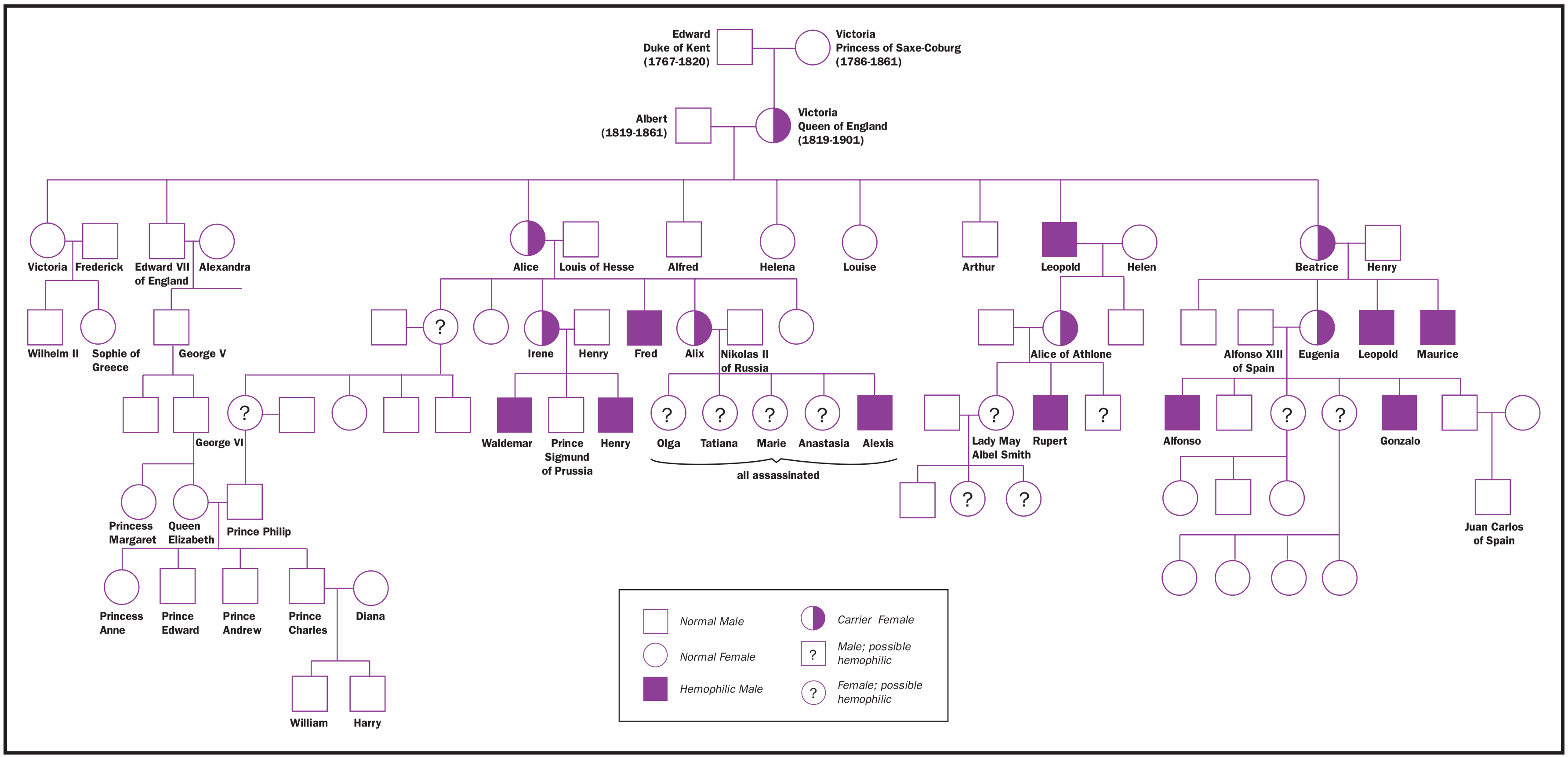
Southern blotting separating DNA fragments by electrophoresis and then identifying a target with a DNA probe

If the gene defect in the family is not known, the inversion mutation in an affected male is generally sought first, as this mutation accounts for at least 20 percent of all cases of hemophilia A, and the test is relatively simple to perform. Although more time-consuming, point-mutation screening also can be done, using a variety of methods. For researchers working on the FVIII or FIX genes, there are sites on the Internet that are valuable resources, as they contain regularly updated listings of all reported mutations in each of these genes, and other useful information.

Because of a high mutation rate, approximately one-third of infants found to have hemophilia A or B have no family history of the disorder, the condition having occurred spontaneously. However, hemophilia is genetically transmitted to future generations.

Differences in Severity of Hemophilia

There are different degrees of severity of hemophilia A and B. Clinical severity usually correlates with the individual's circulating FVIII or FIX level



Pedigree of hemophilia in the royal families of Europe. Queen Victoria was a carrier of this X-linked trait. Her son Leopold inherited the X chromosome that carried it, while her son Edward VII did not. Adapted from http://ublib.buffalo.edu/libraries/projects/cases/hemo.htm.

(determined by doing an FVIII or FIX assay on a venous blood sample). The severity is generally the same in all affected members of a family.

Normal values for FVIII and FIX can range between 50 and 150 percent of the mean value, while severely affected individuals generally have levels of less than or equal to 1 percent, moderately affected persons 1 to 5 percent, and mildly affected persons 5 to 35 percent. Severely affected individuals often have spontaneous joint and muscle hemorrhages, whereas mildly affected persons bleed only with trauma or surgery.

Treatment

Treatment for bleeding episodes consists of replacing the missing clotting factor by intravenous infusion of FVIII (or FIX). There are both human plasma-derived FVIII and FIX concentrates and recombinant DNA-derived FVIII and FIX concentrates. The useful life for both proteins (FVIII and FIX) once infused is relatively short (on average, half is degraded within twelve hours for FVIII and within eighteen hours for FIX). Thus for serious bleeding episodes or surgery, frequent repeat dosing (or continuous infusion) is often necessary.

Prophylaxis is also used, particularly in persons with severe hemophilia A or B. This consists of giving FVIII three times weekly, and FIX twice weekly (in view of its longer half-life, once infused). The aim of prophylaxis (which is often begun between age one and three) is to prevent joint bleeding (and the resulting increase in joint destruction and disability).

Persons with mild hemophilia A can often be treated with the synthetic agent DDAVP (1-deamino-8-D-arginine vasopressin). This analogue of the naturally occurring **antidiuretic** hormone vasopressin results in a rapid release of whatever FVIII (and another large plasma glycoprotein, von Willebrand factor) is in the individual's body storage sites. Thus, following intravenously administered DDAVP, FVIII (and von Willebrand factor) increase (two- to tenfold), but then fall back to baseline within approximately twelve to fifteen hours. This drug comes in several formulations, for **intravenous**, **subcutaneous**, and intranasal use.

antidiuretic a substance that prevents water loss

intravenous into a vein

subcutaneous under the skin

It is hoped that gene therapy for persons with severe hemophilia may eventually become a realistic option. In early 2002 there were several ongoing phase-one trials (very early research studies) in human subjects with severe hemophilia, using different **vectors** and different techniques. However, formidable challenges remain. SEE ALSO BLOTTING; DISEASE, GENETICS OF; GENETIC COUNSELING; INHERITANCE PATTERNS; MUTATION; TRANSPOSABLE GENETIC ELEMENTS; X CHROMOSOME.

vectors carriers

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Bibliography

Lakich, Delis, et al. "Inversions Disrupting the Factor VIII Gene as a Common Cause of Severe Haemophilia A." *Nature Genetics* 5 (1993): 236–241.

Lillicrap, David. "The Molecular Basis of Haemophilia B." *Haemophilia* 4 (1998): 350–357.

Potts, D. M., and W. T. W. Potts. *Queen Victoria's Gene: Haemophilia and the Royal Family.* Gloucestershire, U.K.: Sutton Publishing, 1995.

Internet Resources

Haemophilia B Mutation Database. King's College London. <http://www.kcl.ac.uk/ip/petergreen/haemBdatabase.html>.

Hemophilia: The Royal Disease. University at Buffalo, SUNY. <http://ublib.buffalo.edu/libraries/projects/cases/hemo.htm>.

National Hemophilia Foundation. <http://www.hemophilia.org>.

Heterozygote Advantage

alleles particular forms of genes

Heterozygote advantage is the superior fitness often seen in hybrids, the cross between two dissimilar parents. A heterozygote is an organism with two different **alleles**, one donated from each parent. Fitness means the ability to survive and have offspring. Heterozygote advantage also refers more narrowly to superior fitness of an organism that is heterozygous for a particular gene, usually one governing a disease.

Inbreeding is the practice of repeatedly crossing a single variety of organism with itself, in order to develop a more uniform variety. During this process, the organism becomes homozygous for many genes, meaning that its two gene copies are identical. This is often accompanied by loss of vigor: slower growth, less resistance to disease, and other signs of decreased fitness. This is known as inbreeding depression. Breeding with another variety ("outcrossing") produces offspring that are heterozygous for many genes, and is often accompanied by an increase in size and vigor. This phenomenon had been known to farmers and plant breeders for many years, and was given the name "heterosis" by the U.S. geneticist George Shull in 1916. It is also known as hybrid vigor or, more commonly, as heterozygote advantage.

Agricultural Significance

Hybrid vigor has had a profound impact on agriculture. Yields of hybrid corn are much greater than those of nonhybridized, "open-pollinated" varieties. This of course means that the hybrid "seed corn" must be produced each generation by crossing two distinct, inbred lines. This in turn has given employment to several generations of teenagers who, in summer, detassel corn plants, that is, remove the pollen from one of the parental plants to prevent self-pollination and assure cross-hybridization with the desired variety.

In domesticated animals used for meat or milk production, selection of the best producers to produce high-yielding offspring has been in effect for centuries and has produced dramatic results. In this case, however, essentially all crossing is performed within the same breed and usually involves inbreeding. No serious attempts are made to outcross among different breeds, which would take advantage of heterozygote advantage. Early experiments suggested that outcrossing does not yield favorable results and thus is avoided to this day.

Hypotheses of Heterozygote Advantage

There are two plausible explanations for heterozygote advantage and inbreeding depression, which may both act in a single organism on different genes. No single hypothesis is likely to explain all cases of heterozygote superiority.

Poppy hybrid flowers are larger and more elaborately colored than the parental types. (*Eschscholzia* crossed with *Hailigain*).

The first hypothesis is known as the favorable dominance hypothesis. It is based on the fact that recessive alleles are very often **deleterious** in the homozygous condition, often because recessives code for a defective form of the protein. Thus, possessing at least one dominant allele is favored. Under this hypothesis, the two inbred parents are each homozygous recessive for one or more (different) traits, and each has decreased fitness. Hybridization creates heterozygote offspring that have a dominant (functional) allele for each trait, thus increasing their fitness or vigor. In this hypothesis, heterozygotes are superior to the homozygous recessive condition, but not the homozygous dominant condition.

deleterious harmful

The other explanation for heterosis is that the heterozygote is superior to both homozygotes. This is usually referred to as the overdominance hypothesis. Hybrid vigor in corn is due to overdominance rather than favorable dominance. Overdominance may help explain why harmful recessive alleles remain in the gene pool. Despite the disadvantage of possessing two copies, possessing one copy is advantageous. The molecular explanation for overdominance is less simple, and may be different for different genes (see discussion below).

Heterozygote Superiority in Humans

Is hybrid vigor a common phenomenon in humans? Do traits such as birth weight, height, and other factors improve with outcrossing? In a large, well-designed study of interracial crosses in Hawaii, Newton Morton and his colleagues found no significant beneficial effects among the offspring of an interracial cross when compared to offspring whose parents were from the same racial group. Other studies have found some beneficial effects of outbreeding but, in general, these studies tended to be flawed and are thus unreliable.

In a small number of cases, humans do show a heterozygote advantage, in which the fitness of the heterozygote is superior to either homozygote. The best known example is the β-hemoglobin locus and its relationship to sickle cell disease. Adult hemoglobin is composed of four polypeptides: two α chains and two β chains, coded for by different genes. The β chain is a sequence of 146 amino acids, with glutamic acid in position 6. This normal hemoglobin is referred to as type A. In sickle cell disease, a mutation causes glutamic acid to be replaced by valine at position 6 and is referred to as hemoglobin S. Individuals who are homozygous for the S allele (SS) have sickle cell disease. Untreated, this condition is lethal, and affected individuals do not survive to be old enough to have offspring.

If there were no selective advantage to the recessive allele, we would expect it to slowly be removed from the gene pool, and the frequency of individuals with sickle cell disease should approximate the mutation rate of the normal A allele to the S allele, which is extremely rare. The disease, however, is relatively common in western and central Africa, where the frequency of S allele can be over 15 percent. The reason for this high frequency is due to the heterozygote AS being resistant to the malarial parasite *Plasmodium falciparum.* Thus in areas where malaria is common, AS individuals, who possess one sickling allele, have an advantage over AA individuals, who possess none. Copies of the S allele are lost from the gene pool when they occur in individuals affected with sickle cell disease since they do not reproduce, but more copies are created in the offspring of AS individuals since they are resistant to a severe parasitic disease, namely malaria. This advantage compensates for the loss of individuals with sickle cell disease, and therefore keeps the S gene at a relatively high frequency in western and central African populations. Other hemoglobin abnormalities, including hemoglobin C and E, also seem to have the same effect as sickle cell disease, though the effect is not as pronounced.

Thalassemia is another hemoglobin disorder that causes severe anemia. There are two types, α and β, which are due to mutations at the α and β hemoglobin loci respectively. The disease has its highest frequency in areas bordered by the Mediterranean Sea, especially Sardinia, Greece, Cyprus, and Israel. Similar to sickle cell disease, the disorder is fatal at an early age and the heterozygotes are resistant to malaria.

The same is true for a deficiency of the enzyme glucose-6-phosphate dehydrogenase (G6PD) deficiency, which has a similar geographic distribution as thalassemia. The disease is X-linked, thus affecting boys. Its persistence in this case is due to heterozygous females being resistant to malaria. Other proposed cases of heterozygote superiority in humans are more speculative, and have not been confirmed by repeated studies. SEE ALSO HARDY-WEINBERG EQUILIBRIUM; HEMOGLOBINOPATHIES; INHERITANCE PATTERNS; POPULATION GENETICS.

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Bibliography

Cavalli-Sforza, L. L., and W. F. Bodmer. *The Genetics of Human Populations.* Mineola, NY: Dover Publications, 1999.

Hartl, D. L., and A. G. Clark. *Principles of Population Genetics,* 3rd ed. Stamford, CT: Sinauer, 1997.

Li, Ching Chun. *First Course in Population Genetics.* Pacific Grove, CA: Boxwood Press, 1976.

Morton, Newton E., Chin S. Chung, and Ming-Pi Mi. *Genetics of Interracial Crosses in Hawaii.* New York: S. Karger, 1967.

High-Pressure Liquid Chromatography *See HPLC: High-Performance Liquid Chromatography*

High-Throughput Screening

High-throughput screening (HTS) is an automated method for rapidly analyzing the activity of thousands of chemical compounds. It has become a key tool in modern drug discovery. Paired with combinatorial chemistry and **bioinformatics**, HTS allows potential drugs to be quickly and efficiently screened to find candidates that should be explored in more detail.

bioinformatics use of information technology to analyze biological data

How the Process Works

Most drugs work by binding to a protein target on or in a living cell. One of the first steps in drug discovery and development is finding molecules that will bind to the target. Imagine, for instance, you want to develop an anticancer drug that binds to and inactivates a particular mutant protein known to promote aberrant cell growth. You have a couple of compounds that bind very weakly to your protein, and these serve as the starting point for generating a large number of related compounds, through combinatorial chemistry. In this method, many thousands of related compounds can be quickly and automatically synthesized. Those that bind best can be modified and tested further, and ultimately may go on to be tested in animals and people as candidate drug therapies.

Initial screening of these compounds for their binding ability is the job for HTS. The key to HTS is to develop a test, or assay, in which binding between a compound and a protein causes some visible change that can be automatically read by a sensor. Typically the change is emission of light by a **fluorophore** in the reaction mixture. One way to make this occur is to attach the fluorophore to the target protein in such a way that its ability to fluoresce is diminished (quenched) when the protein binds to another molecule. A different system measures the difference in a particular property of light (polarization) emitted by bound versus unbound fluorophores. Bound fluorophores are more highly polarized, and this can be detected by sensors. Other detection methods are possible as well.

fluorophore fluorescent molecule

The details of HTS differ with different systems, but all depend on automated or robotic systems to combine the chemicals and read the outputs. Reactions between the target protein and the compound usually occur in microplates, which are plastic trays with multiple indentations, or wells. Systems currently in use can handle plates with 96, 384, 1,536, or even higher numbers of wells at once. HTS typically uses extremely small volumes in each well, often 10 microliters or less. Small volumes have numerous advantages, including keeping to a minimum the amount of each compound used. This is especially important for many proteins targets, which may be difficult and costly to isolate and purify.

The time required for reactions varies with the substances involved, and may range from several minutes to several hours. Fast robotic systems combined with rapid reactions can screen 10,000 or more compounds in a single day. This is an enormous increase over traditional chemical assays, in which a chemist may be able to handle fewer than 100 tests in the same amount of time.

The Uses of HTS Assays

Storing, processing, analyzing, and accessing the wealth of data generated in an HTS assay poses special problems, simply because there is so much of it. Bioinformatics strategies are used to develop databases relating chemical structure, target characteristics, and assay results, allowing researchers to learn more from their results than just whether or not a particular compound was successful. Analyzing the common features of successful compounds may lead to rational development of better drug candidates.

High-throughput technology can also be put to use in other areas besides drug development. Indeed, any system in which there are many similar candidates to be screened, and in which a visible output can be designed, is amenable to high-throughput methods. **Genomics** applications are a principal area for applying HTS technology, in DNA sequencing, protein analysis, and other fields. HTS methods can be combined with DNA microarray technology, for instance, to analyze the expression of hundreds of different genes under varying conditions. SEE ALSO Bioinformatics; Biotechnology; Combinatorial Chemistry; DNA Microarrays; Proteomics.

genomics study of gene sequences

Richard Robinson

Bibliography

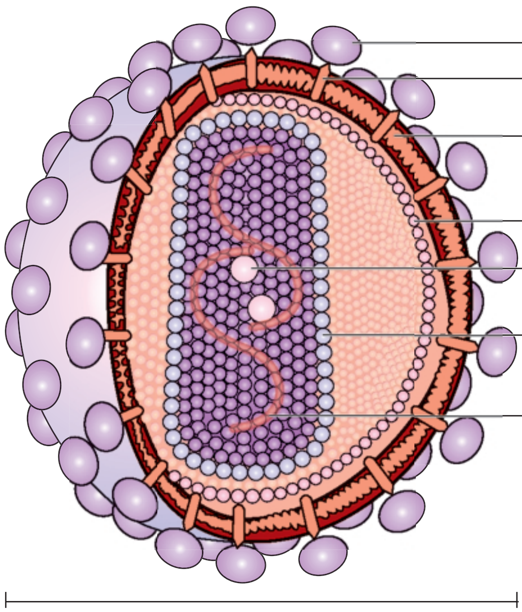
Brush, Michael. "High-Throughput Technology Picks Up Steam." *Scientist* 13, no. 4 (February 15, 1999): 11.

Internet Resource

High Throughput Screening. <http://www.htscreening.net/>.

HIV

HIV, the human immunodeficiency virus, is the virus that causes AIDS, a debilitating and deadly disease of the human immune system. HIV is one of the world's most serious health problems: at the end of 2001, more than 40 million people worldwide were infected with HIV and living with the virus or AIDS. The World Health Organization estimates that about 20 million people have died from AIDS since the infection was first described in 1981. Nearly 500,000 of those deaths have occurred in the United States. Although there is no cure for the disease, therapies exist that reduce the



The HIV virion (virus particle) has two glycoproteins on the outside, which aid its passage into human cells. It contains two copies of its RNA genome, plus the reverse transcriptase enzyme used to copy the RNA into DNA once it is inside the host cell.

symptoms of AIDS and can extend the life spans of HIV-infected individuals. Researchers are also pursuing protective vaccines, but a reliable vaccine might still require years to develop.

HIV and AIDS

HIV infects certain cells and tissues of the human immune system and takes them out of commission, rendering a person susceptible to a variety of infections and cancers. These infections are caused by so-called opportunistic agents, **pathogens** that take advantage of the compromised immune system but that would be unable to cause infection in people with a healthy immune system. Rare cancers such as Kaposi's sarcoma also take hold in HIV-infected individuals. The collection of diseases that arise because of HIV infection is called acquired immune deficiency syndrome, or AIDS. HIV is classified as a lentivirus ("lenti" means "slow") because the virus takes a long time to produce symptoms in an infected individual.

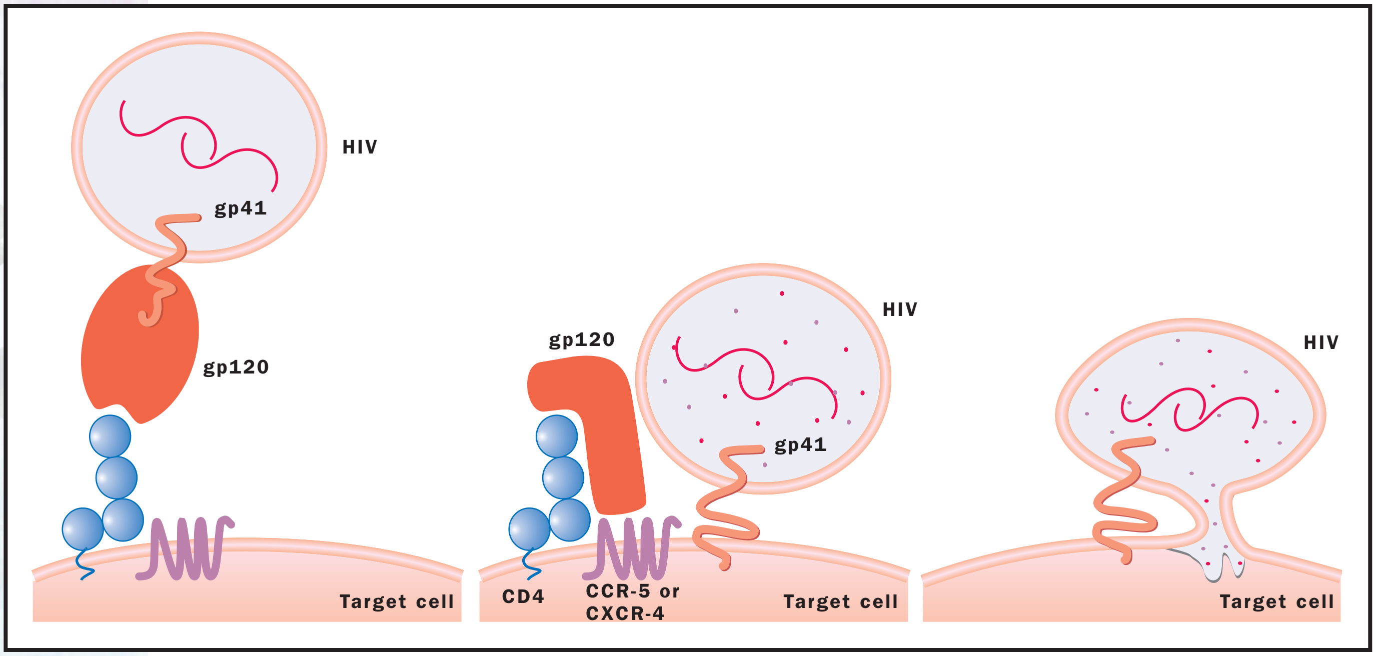
pathogens disease-causing organisms

HIV Life Cycle: Entering Cells

Like a typical virus, HIV infects a cell and appropriates the host's cellular components and machinery to make many copies of itself. The new viruses then break out of the cell and infect other cells. HIV stores its genetic information on an RNA molecule rather than a DNA chromosome. This is a distinguishing characteristic of **retroviruses**, which are viruses that must first convert their RNA genomes into DNA before they can reproduce.

retroviruses RNA-containing viruses whose genomes are copied into DNA by the enzyme reverse transcriptase

Each HIV virion (viral particle) is a small sphere composed of several layers. The external layer is a membrane coat, or envelope, obtained from the host cell in which the particle was made. Underneath this membrane lies



HIV's gp120 protein interacts with two proteins on the surface of the target cell. Gp41 then promotes fusion of the virus membrane with the cell membrane, allowing the viral contents to enter the cell. Adapted from an article in *Molecular Medicine Today,* 1998.

antibodies immune-system proteins that bind to foreign molecules

a shell made from proteins, called a nucleocapsid. Inside the protein shell are two copies of the virion's RNA genome and three kinds of proteins, which are used by the virion to establish itself once inside the cell that it infects.

Two proteins, called gp120 and gp41, enable the virion to recognize the type of cell to enter. These proteins project from the HIV membrane coat. Gp120 binds to two specific proteins found on the target cell's surface (these target-cell proteins are called receptors). The first receptor, CD4, is found on immune system cells known as CD4 T cells, and also sometimes on two cell types known as macrophages and dendritic cells. The immune system uses CD4 T cells in the initial step in making **antibodies** against infectious agents. After binding to CD4, the HIV protein called gp120 binds with a second cell membrane protein, commonly referred to as the co-receptor. The co-receptor can be one of many different proteins, depending on the cell type. The two most common are CXCR4, which is normally found on CD4 T cells, and CCR5, a receptor found on CD4 T cells as well as on certain macrophages and dendritic cells. In the absence of HIV, CXCR4 and CCR5 allow these immune system cells to respond to chemical signals, but when HIV infects the cells, the HIV commandeers their usage. In some cases, individuals have a mutation in their co-receptor that prevents HIV from entering their cells.

Once gp120 has bound to both the CD4 receptor and co-receptor, the gp41 protein fuses HIV's membrane envelope with the cellular membrane, injecting the virus into the target cell. Once in the cytoplasm, the viral protein shell opens up and releases the viral proteins—a reverse transcriptase, a viral integrase, and a protease—along with the viral RNA strands. The reverse transcriptase copies the RNA strands into DNA. The viral integrase then helps insert the DNA copies into the cell's chromosome. At this point, the virus is called a provirus, and the life cycle halts. The provirus may remain dormant in the cell's chromosome for months or years, waiting for the T cell to become activated by the immune system.

HIV Life Cycle: Reproduction

When the immune system recruits T cells to fight an infection, the T cells start producing many proteins. Along with the normal cellular protein products, a T cell carrying an HIV provirus also produces HIV proteins. The first HIV proteins made are called Tat and Rev. Tat encourages the cellular machinery to copy HIV's proviral DNA into RNA molecules. These RNA molecules are then processed in the nucleus to become templates for several of the HIV proteins, some of whose functions are not well understood.

Rev, on the other hand, ushers the HIV's RNA molecules from the nucleus, where they are being reproduced, into the host cell's **cytoplasm**. Early in HIV reproduction, with only a few RNA molecules from which to make protein, a small quantity of Rev is made. Therefore, most of the RNA molecules remain in the nucleus long enough to get processed. As time passes, however, and Tat continues to instigate RNA production, more Rev is made. A higher amount of Rev protein increases the speed with which RNA molecules are ejected from the nucleus. These RNA molecules, which have undergone little or no processing, become templates to make different HIV proteins. The newer proteins are made in long chains that require trimming before they become functional. One of the proteins in the chain is the protease, the protein that trims. Other proteins include those that make up the protein shell, the reverse transcriptase, and integrase.

cytoplasm the material in a cell, excluding the nucleus

After the newly created proteins are processed to the right size, they form new virions by first assembling into a shell, then drawing in two unprocessed RNA molecules and filling up the remaining space with integrase, protease, and replicase. The new virions bud from the host-cell membrane, appropriating some of that membrane to form an outer coat in the process. The mature virus particles are now ready to infect other cells.

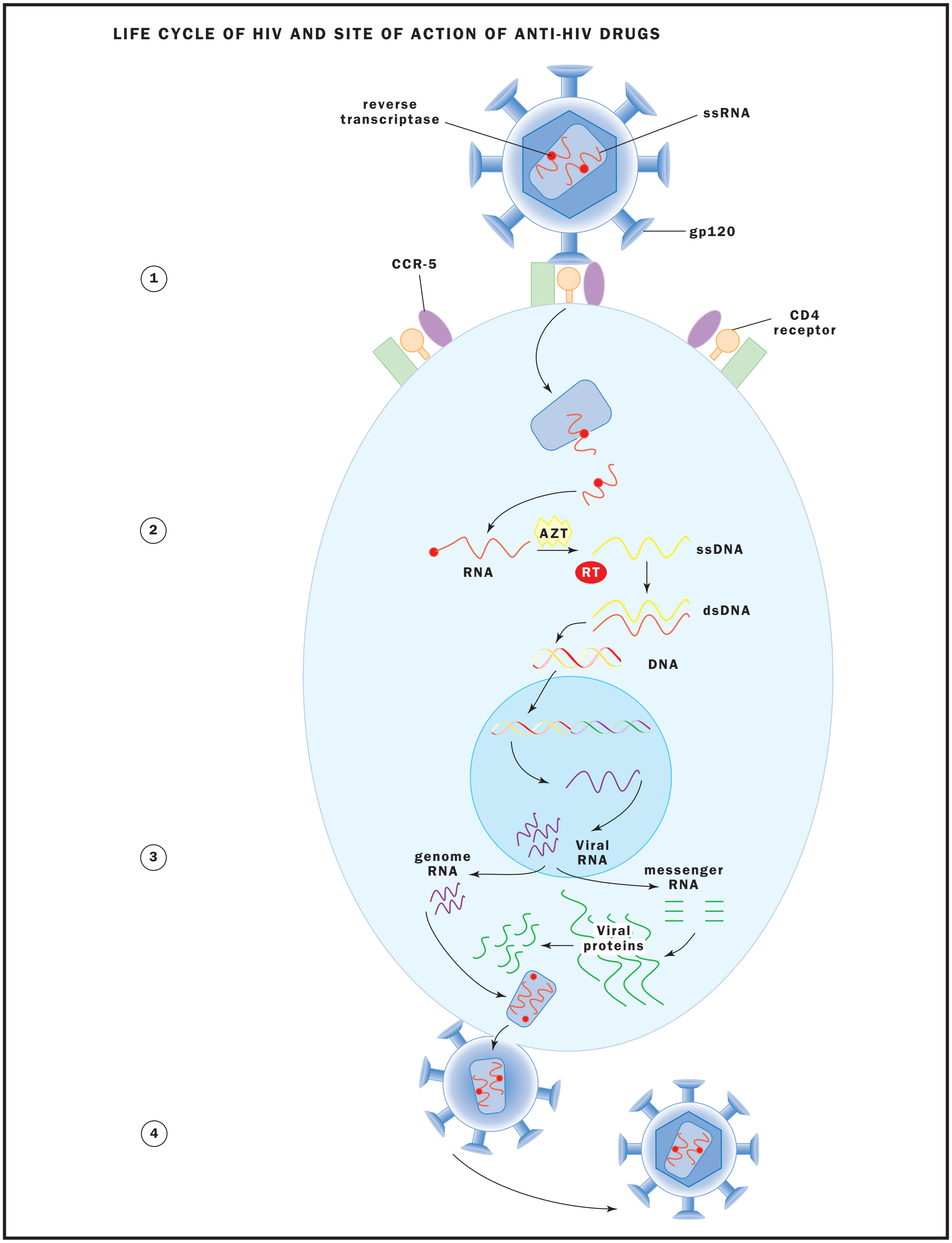
HIV's Immune-System Impairment Mechanism

One of the most disastrous effects of HIV infection is the loss of the immune system's CD4 T cells. These cells are responsible for recognizing foreign invaders to a person's body and initiating antibody production to ward off the infection. Without them, people are susceptible to a variety of diseases. HIV destroys the T cells slowly, sometimes taking a decade to destroy a person's immunity. However, in all the time before an HIV-infected individual shows any symptoms, the virus has been reproducing rapidly. The lymph tissue, the resting place for CD4 T cells, macrophages, and dendritic cells, becomes increasingly full of HIV, and viral particles are also released into the bloodstream.

HIV's main target is the population of CD4 T cells within a host's body. HIV kills them in one of three ways. It kills them directly by reproducing within them, then breaking them upon exit; it kills them indirectly by causing the cells to "commit suicide" by inducing **apoptosis**; or it kills them indirectly by triggering other immune cells to recognize the infected T cell and kill it as part of the immune system's normal function.

apoptosis programmed cell death

As infected T cells die, the immune system generates more to take their place. As new T cells become infected, they are either actively killed or induced to commit suicide. Meanwhile, the HIV virus is not completely



HIV binds to CD4 and CCR-5 or CXCR4 on the surface of the target cell, allowing the virus coat to fuse with the cell membrane and virus contents to enter the cell. (2) Reverse transcriptase (RT) enzyme transcribes viral RNA into single-strand viral DNA, a step blocked by RT inhibitors (AZT, ddi, d4t, ddc). Single-strand DNA is converted to double-strand DNA which then enters the nucleus and integrates into the host chromosome. (3) Upon activation of the cell, viral RNA triggers manufacture of viral proteins. In the absence of protease inhibitors, proteins are processed into shorter lengths. Protease inhibitors (saquinavir, ritonavir, indinavir, nelfinavir) prevent this step, interrupting the viral life cycle. (4) After all components are present in the cell, HIV particles assemble and bud from the cell. Adapted from Stine, 1997.

hidden from the immune system. As with any infectious agent, HIV presents its proteins to the immune system, which develops antibodies against it. This antibody production, however, is hampered by the fact that HIV mutates rapidly, changing the proteins it displays to the immune system. With each new protein, the immune system must generate new antibodies to fight the infection. Thus, an HIV infection is a dramatic balance between a replicating, ever changing virus and the replenishing stores of T cells that are fighting it. Unfortunately, the immune system, without therapeutic intervention, eventually loses the battle.

Once the CD4 T cells are depleted, the immune system can no longer ward off the daily bombardment of pathogens that all human organisms experience. Common infectious agents thus overwhelm the system, and HIV patients become susceptible to a variety of "opportunistic" diseases that take advantage of the body's reduced ability to fight them off. AIDS doctors report at least twenty-six different opportunistic diseases specific to HIV infection. These include unusual fungal infections such as thrush. The chickenpox virus may come out of dormancy, manifesting itself as the painful disease known as shingles. An obscure form of pneumonia, called pneumocystis pneumonia, is also common in AIDS patients. In addition, patients can acquire cancers such as B-cell lymphoma, which is a cancer of the immune system. Doctors generally consider patients with fewer than 200 CD4 T cells per cubic milliliter of blood as having AIDS. (In contrast, a healthy person counts more than 1,000.)

Anti-HIV Drug Therapy

Drugs that interfere with viral replication can slow down HIV disease. Early trials relied on the administration of one drug at a time. While patients' health improved and their T cell count rose, in time HIV mutated enough to render the drugs ineffective. Since 1995, however, doctors have found that rotating patients through three different drugs in very high doses significantly improves the health of AIDS patients. Known as "highly active antiretroviral therapy" (HAART), this therapeutic approach also reduces the amount of HIV circulating in the bloodstream to nearly undetectable levels. People infected with HIV who are treated by HAART are now living longer, healthier lives than ever before.

Targeting Life-Cycle Points

Drugs meant to knock out HIV target the activities of two HIV proteins, the reverse transcriptase and the protease. HAART requires drugs of both types. Drugs called protease inhibitors prevent the viral protease from trimming down the large proteins made late during infection. Without those proteins, the viral shell cannot be assembled. In addition, the proteins that reproduce HIV's genetic information, the reverse transcriptase and the integrase, are not functional.

Drugs that inhibit the reverse transcriptase prevent it from copying the RNA into DNA. These drugs work early in the life cycle of HIV. Reverse transcriptase inhibitors include azidothymidine (AZT), whose structure resembles the DNA nucleotide thymine. When reverse transcriptase builds DNA with AZT instead of thymine, the AZT caps the growing DNA molecule and halts DNA production, due to AZT's slight difference in structure from the thymine that DNA production requires. SEE ALSO IMMUNE SYSTEM GENETICS; RETROVIRUS; REVERSE TRANSCRIPTASE.

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Bibliography

Janeway, Charles A., et al. "Failures of Host Defense Mechanisms." In *Immunobiology: The Immune System in Health and Disease*, 4th ed. New York: Current Biology Publications, 1999.

———. *HIV Infection and AIDS: An Overview*. Washington DC: National Institute of Allergy and Infectious Diseases and U.S. Department of Health and Human Services, 2001.

Shilts, Randy. *And the Band Played On: Politics, People, and the AIDS Epidemic*. New York: St. Martin's Press, 2000.

Stine, Gerald. *Acquired Immune Deficiency Syndrome: Biological, Medical, Social and Legal Issues*, 3rd ed. New York: Prentice-Hall, 1997.

Homology

Homology is used to describe two things that share a common evolutionary origin. In genetics and molecular biology, homology means that the sequences of two different genes or two different proteins are so similar that they must have been derived from the same ancestral gene or protein.

The word "homology" has several meanings in biology, each related to the word's origin, meaning "same knowledge." At a molecular level, the term "homology" describes sequences, either DNA or protein, that share a common evolutionary origin. On a larger scale, a pair of chromosomes from a **diploid** organism that have the same size and shape, are considered homologous chromosomes. Regions of each member of a chromosome pair, which carry the same set of genes, are homologous regions. Finally, physical features with a common evolutionary origin, such as the wing of bat and the hand of a human, are homologous structures.

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

Diversity and Natural Selection

Biologists have long been fascinated by the diversity of life. The amazing variety of living things makes it natural to wonder how so many different life-forms came to be. Physical characteristics that could be easily observed, such as the shape of wing, the structure of a shell, or the size of a beak, provided the first means to search for an answer. Recognition of the variation within a species (imagine a Chihuahua and a Great Dane) led Charles Darwin to propose that new species emerge when selection favors certain traits within a population.

Today's biologists continue to study the effects of natural selection on the evolution of species, but they are no longer limited to beak size and wing shape. Now they can compare the positions of genes on chromosomes, the amino acid sequences of proteins, and the nucleotide sequences of genes. With DNA or protein sequences from over 133,000 species represented in the taxonomy database at the National Center for Biotechnology Information (NCBI) and over 800 **genome** sequences either published or in progress, researchers have an unprecedented opportunity to study evolution at a molecular level.

genome the total genetic material in a cell or organism